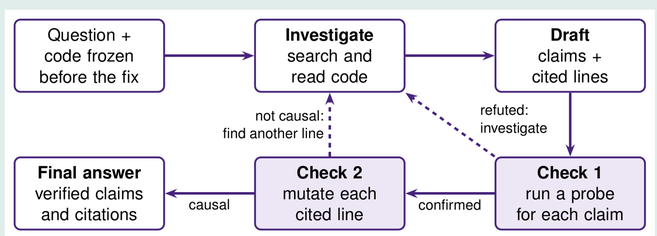
Run It, Then Cite It: Execution- and Mutation-Verified Answers to Developer Questions

ABSTRACT

Developers often ask how a library behaves when its documentation may be inaccurate. LLM assistants tend to repeat the documentation and cite code that does not determine the behaviour. We propose an agent that verifies its answers in two ways: it executes a test for each claim, and it accepts a cited code line only if modifying that line changes the test outcome. We evaluate it on 145 GitHub issues from click and marshmallow against a ReAct agent with the same tools and budget. The verification procedure did not improve accuracy, and the line-modification check proved unreliable on real code.

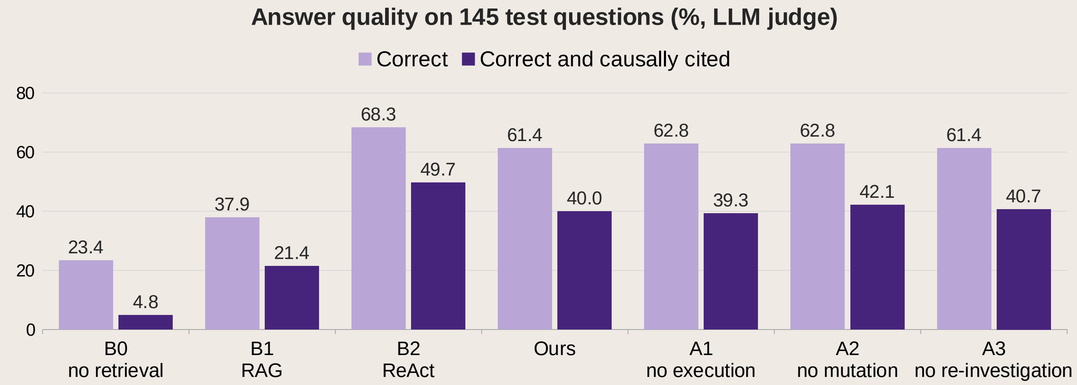
METHODOLOGY

Dataset. 170 questions built from GitHub issues, following SWE-bench: the merged fix and its test form the ground truth. The library is frozen at the commit before the fix. After a human quality check, 145 test questions remain.
Agent. The agent drafts an answer as claims with cited code lines. Check 1 executes a test (probe) for each claim in an isolated sandbox. Check 2 modifies each cited line and accepts it as causal only if the claim's outcome changes while control tests stay the same.
Baselines. B0: no retrieval. B1: retrieval, single call. B2: ReAct agent with the same tools and budget.



RESULTS

On 145 test questions, our agent did not outperform the ReAct baseline (B2): 40.0% vs 49.7% of answers were both correct and causally cited (LLM judge), and 25.0% vs 41.7% on 12 human-labelled questions. Both tool-using agents clearly outperform the baselines without tools.



HYPOTHESES

All hypotheses were pre-registered. Human labels are the primary measure, because the LLM judge showed low agreement with human annotators (kappa 0.36).

	Hypothesis	Result
H1	Our agent outperforms B2 by at least 8 points	Not supported (25.0% vs 41.7%)
H2	At least 40% fewer confident errors than B2	Not supported (29.7% vs 31.7%)
H3	Mutation check raises citation precision by at least 15 points	Not supported (30.6% vs 44.1%)

MUTATION CHECK ANALYSIS

The acceptance rule used by the agent accepted only 24% of the lines changed in the developers' fixes, and its agreement with human judgements was at chance level. Rejecting correct lines led the agent to replace good citations with weaker ones.

Acceptance rule	Fix lines accepted	Agreement with humans (kappa)
Any outcome change	66.8%	0.20
Behavioural change	56.1%	0.17
Specific change (used)	24.0%	−0.10

Disco 82

LMA Group Project

CONCLUSION AND FUTURE WORK

- Execution- and mutation-based verification did not outperform a ReAct agent with the same tools.
- Mutation is not a reliable test of causality in real code.
- Flagging failed checks, instead of re-investigating them, reduced confident errors by one third.
- No unsafe code was executed in any experiment.
- Future work: a graded causal test and a Chain-of-Verification baseline.

ACKNOWLEDGMENT

We thank the instructors and teaching assistants of CL3.410. Experiments used the free tiers of Google Gemini and Groq.

REFERENCES

[1] Yao et al. ReAct. ICLR 2023.
[2] Jimenez et al. SWE-bench. ICLR 2024.
[3] Dhuliawala et al. Chain-of-Verification. ACL Findings 2024.
[4] Moon et al. Ask the Mutants (MUSE). ICST 2014.
[5] Kiecker et al. CASCADE. FSE 2026.

Code, data and full report
github.com/ameya-purohit/LMA-Final-Project

Authors: Ameya Purohit, Sunil Kumar, Aditya Narayan Dwivedi, Sourav Nath

Course: CL3.410, Monsoon 2026